

A proper spectral modification violating Efimov rigid-fibre invariance

Theorem. *Let*

$$k = \mathbf{Q}, \quad R = H(k[[t]]), \quad J = \bigoplus_{n \geq 1} \Sigma^n(R/t^n), \quad A = R \oplus J,$$

where A is the split square-zero E_∞ -extension, and put $K = R[1/t] \simeq A[1/t]$. Then $\mathrm{Spec} R \rightarrow \mathrm{Spec} A$ is proper and locally almost of finite presentation and is an equivalence over $D(t)$, but the canonical pullback

$$\mathrm{Rig}(\mathrm{Mod}_A^{t\text{-comp}}) \otimes_{\mathrm{Mod}_A} \mathrm{Mod}_K \longrightarrow \mathrm{Rig}(\mathrm{Mod}_R^{t\text{-comp}}) \otimes_{\mathrm{Mod}_R} \mathrm{Mod}_K \quad (1)$$

is not fully faithful.

Proof. Every bounded truncation of J is a finite sum of perfect R -modules, so $J \in \mathrm{APerf}(R)$; the square-zero criterion makes $A \rightarrow R$ locally almost of finite presentation. Its classical truncation is the identity, hence the induced spectral morphism is proper, and $J[1/t] = 0$ gives $A[1/t] \simeq R[1/t]$.

Completed scalar extension and restriction give

$$\Phi(M) = \Lambda_t(R \otimes_A M) : \mathrm{Mod}_A^{t\text{-comp}} \rightleftarrows \mathrm{Mod}_R^{t\text{-comp}} : \mathrm{Res},$$

and hence an adjunction $F \dashv G$ on rigidifications. Let

$$a_A : \mathrm{Rig}(\mathrm{Mod}_A^{t\text{-comp}}) \rightarrow \mathrm{Mod}_A^{t\text{-comp}}, \quad a_R : \mathrm{Rig}(\mathrm{Mod}_R^{t\text{-comp}}) \rightarrow \mathrm{Mod}_R^{t\text{-comp}}$$

be the standard maps from rigidification, and let Λ_A, Λ_R be their fully faithful symmetric-monoidal right adjoints. Functoriality and passage to right mates give

$$a_R F \simeq \Phi a_A, \quad G \Lambda_R \simeq \Lambda_A \mathrm{Res}. \quad (2)$$

The usual unit-and-mate calculation identifies the unit of $F \dashv G$ at the tensor unit with $\Lambda_A(A \rightarrow R)$; consequently

$$\mathrm{fib}(\mathbf{1} \rightarrow GF(\mathbf{1})) \simeq \Lambda_A(J). \quad (3)$$

After tensoring with K , its fibre is

$$L_t \Lambda_A(J) := \mathrm{colim}(\Lambda_A(J) \xrightarrow{t} \Lambda_A(J) \xrightarrow{t} \cdots). \quad (4)$$

It remains to prove that (4) is nonzero.

Set $Q = \bigoplus_{m \geq 1} \Sigma^{-m} R$, with A -action through $A \rightarrow R$. The shifts make J, Q derived t -complete, and both are ω_1 -compact in $\mathrm{Mod}_A^{t\text{-comp}}$. Put $\mathcal{D} = (\mathrm{Mod}_A^{t\text{-comp}})^{\omega_1}$. Efimov's predual description gives, for the compact representable $Q^{op} \in \mathrm{Ind}(\mathcal{D}^{op})$,

$$\mathrm{Map}(Q^{op}, \Theta_A L_t \Lambda_A(J)) \simeq \mathrm{Map}_{\mathrm{Mod}_A^{t\text{-comp}}}(\mathbf{1}, J \widehat{\otimes}_A Q)[1/t], \quad (5)$$

where $\Theta_A : \mathrm{Rig}(\mathrm{Mod}_A^{t\text{-comp}}) \hookrightarrow \mathrm{Ind}(\mathcal{D}^{op})$ is the canonical fully faithful predual realization. Thus it suffices to prove that the right side of (5) is nonzero.

Because the A -actions on J, Q factor through R ,

$$J \otimes_A Q \simeq J \otimes_R (R \otimes_A R) \otimes_R Q. \quad (6)$$

In $\mathrm{Mod}_{R \otimes R^{op}}$, the R -bimodule R is a retract of $R \otimes_A R$. Hence $J \otimes_R Q$ is a retract of $J \otimes_A Q$. Its equal-index summands give a further retract

$$D := \bigoplus_{n \geq 1} R/t^n \hookrightarrow J \otimes_A Q, \quad (7)$$

so $\Lambda_t D$ is a retract of $J \widehat{\otimes}_A Q$.

For $r \geq 1$, define

$$x^{(r)} = (t^{\lfloor n/2 \rfloor} \bmod t^{\min(n,r)})_{n \geq 1} \in \pi_0(D \otimes_R^{\mathbf{L}} R/t^r).$$

For fixed r this has finite support; the $x^{(r)}$ are compatible, and no power of t kills their inverse-limit class. The Milnor exact sequence surjects

$$\pi_0(\Lambda_t D) \twoheadrightarrow \varprojlim_r \pi_0(D \otimes_R^{\mathbf{L}} R/t^r),$$

so $(\Lambda_t D)[1/t] \neq 0$. By (7), the right side of (5) is nonzero; hence $L_t \Lambda_A(J) \neq 0$. Therefore the localized adjunction unit is not invertible, and (1) is not fully faithful. $\square$

References: A. I. Efimov, *Localizing invariants of inverse limits*, arXiv:2502.04123v2, Prop. 1.27, Prop. 1.34, Thm. 4.2 and Cor. 4.4; the standard square-zero criterion for locally almost finitely presented spectral algebras.